



Federal Bank: Humanizing the banking experience with Vertex AI

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GOOGLE CLOUD RESULTS

- ✓ Transforms the banking experience with chatbot driven by Vertex AI for more seamless and relevant content
- ✓ Automate workflows with API Support Portal built with Google Cloud APIs
- ✓ Enables easy app integration and insights with Firebase and BigQuery
- ✓ First bank in India to implement gen AI across its platform

Federal Bank is humanizing the banking experience and providing customers with more seamless access to information they need with Vertex AI.

To make banking a more satisfying experience for customers, banks around the world have been undergoing digital transformation. For [Federal Bank](#), which has a mission to be customer-centric with speedy and targeted services, improving its existing Virtual Assistant capabilities was a priority, while it continued to modernize its infrastructure for better integration with fintech partners to expand its service offerings.

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Group President and
CFO, Federal Bank



In early 2022, Federal Bank started working with [Google Cloud](#), tapping onto its [Dialogflow](#) and [Vertex AI](#) capabilities to elevate the capabilities of its chatbot, Feddy. In particular, the bank wanted to humanize its interactions with customers so that it could provide a more personalized experience at every touchpoint.

Federal Bank also looked to develop a mobile app for employees so that they could work more efficiently, and leveraged [Firebase](#) due to its fully managed infrastructure, and easy integration with the Google Cloud ecosystem.

All of these were built on Google [Compute Engine](#) and with APIs as a key part of its infrastructure, the bank could easily integrate with organizations outside of its ecosystem to broaden its service offerings.



Improving chatbot features with natural language responses

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Saran Joseph

When Federal Bank first deployed Google Cloud in early 2022, the team was focused on improving its chatbot, Feddy. It used Dialogflow, specifically Dialogflow CX, to translate text input from Feddy into structured data for processing before churning out a response in a conversational tone. Dialogflow enabled Feddy to answer customer queries and manage basic transactions.

However, when [Gemini](#) was introduced in December 2023, the bank decided to upgrade its features and functionality to support even more seamless conversations with customers. Beyond improving Feddy, the bank implemented the same

Vice President, Federal
Bank

generative AI technology on its search bar on the website so that customers could ask questions and be provided with human-like responses, instead of being served with links that may or may not be directly applicable to them.

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"Of course, the AI-equipped chatbots are not going to replace humans," Joseph says, "If customers are not comfortable liaising with the AI platform alone and want to speak to a person, there is still that option for them to do so."



Leveraging APIs for deeper collaboration and growth

Apart from modernizing its chatbot, Federal Bank developed its own API support portal, which is an API integration tool with [Google Cloud APIs](#) to automate its workflows and better collaborate with fintech organizations. The API support portal lets engineers seamlessly generate sample codes in Python or Java or any other coding language, depending on the parties involved. It can also debug issues that come about, such as if backend APIs are not responding, so the IT support team at Federal Bank can be freed from resolving minor issues that can be automated.

"Google Cloud APIs has helped us improve security and enable faster go-to-market as we have the flexibility to adapt and scale our systems with ease, ensuring interoperability across various platforms and services," says Joseph. The user experience on the platform has also improved, while the data team has been able to deliver deeper analytics.

With its employee mobile app built on [Firebase](#), the team leverages Firebase Analytics and [BigQuery](#) as its data warehouse to better understand app user habits for continuous improvement. For

example, the team can gain visibility of the full customer journey within the app, such as how many times they visit the app, the number of clicks, and which functionality is most used. With these insights, the development team can improve its application over time for an even more seamless user experience.



Safeguarding data to the highest standards

As a bank, security is always a priority, and for that, the bank leverages [Cloud Armor](#) to protect its app and website against distributed denial of service (DDoS) attacks. But more importantly, it is the promise of Google Cloud that makes it a trusted organization for the bank.

"All data that resides in Google Cloud are not cross-utilized, which means it is not used in any way to train the foundational model. This gives us the peace of mind knowing that our customers' data is always safe, and also provides a sense of security to our customers," shares Joseph.



Innovating alongside a dedicated team

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While the technology that Google Cloud provides is critical to the bank’s success, Joseph shares that he appreciates having a dedicated customer engineering support and principal architects support team to oversee the Google Cloud implementation.

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Johnson K Jose
Group President and
CTO, Federal Bank

bolster security measures, and introduce innovative products that redefine the banking experience for our customers. Leveraging Google's Vertex AI, we aim to deploy intelligent search bots that significantly reduce response times and enable us to handle greater volumes with existing resources. Together, we are shaping the future of banking through AI innovation," says Johnson K Jose, Group President and CTO, Federal Bank.

Moving forward, Joseph and his team are confident they would continue investing in Google Cloud. With more than 90 percent of their customer base using Android devices, it makes sense from a business perspective to look at the One Google ecosystem, as well as to provide a better customer experience as a whole.

"Federal Bank and Google Cloud have embarked on a strategic partnership to co-create and co-innovate and reimagine the future of banking. By harnessing the power of Vertex AI we've developed an intelligent Gen AI search bot that provides 24/7 support to our customers. We are also exploring new possibilities in personalized

financial recommendations, increased sales conversations, fraud detection, and hyper-efficient operations," says Joseph.

Headquartered in Aluva, Kochi, Kerala, **Federal Bank** is a leading private sector bank with 1,408 outlets and more than 16 million customers across India. Its services include B2B integrations as well as a retail segment.

Industries: Financial Services

Location: India

Products: [Google Cloud](#), [BigQuery](#), [Dialogflow](#), [Firebase](#), [Migrate for Compute Engine](#), [Natural Language AI](#), [Vertex AI](#), [Vertex AI Notebooks](#)

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